

FT8-Helper-3.0.1 User Guide

Since the functions of WSJT-X and JTDX have been divided, and the further development of JTDX has come to a standstill, JTDX is no longer supported by the FT8-Helper. The last version of JTDX-Helper is still available at www.github.com/dg5lp/JTDX-Helper-160.

The FT8-Helper (hereinafter "Helper") was developed as a macro extension using the Quick-Macros program. The Helper interprets the data received from WSJT-X and acts intelligently according to various selectable strategies. The Helper does not require a license for Quick-Macros to run. (Quick Macros © Gintaras Didzgalvis).

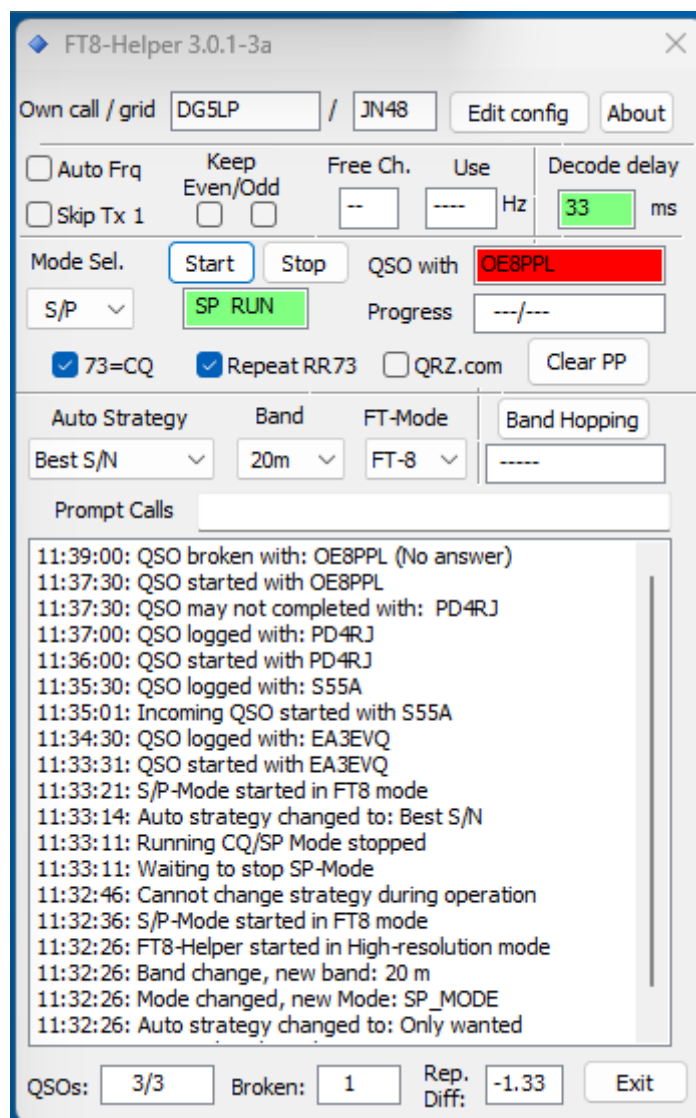
The Helper controls WSJT-X via virtual mouse and keyboard commands. When the Helper is running, you should not use your computer for other applications, as the mouse focus is intercepted by the Helper every 7.5 or 15 seconds depending on the used FT4/FT8 mode.

Please note that the main version number of WSJT-X (here 3.0.1) must match the version number of the helper. The number after "3.0.1-" is the consecutive version number of the Helper.

It should also be noted that currently the "Enhanced" versions of WSJT-X are not supported by the Helper.

Main Features of FT8-Helper-3.0.1

- Automatic operation in both "CQ" and "S/P" mode.
- Automatic detection of a clear operating frequency in CQ- an MIX mode
- Automatic switching between CQ and S/P mode with programmable intervals, depending on band activity.
- In S/P-mode the Helper can call stations transmitting 73 or RR73.
- A wide range of filtering strategies available:
 - Depending on the strategy, stations are called based on the following criteria:
 - Those with the highest priority, in accordance with the "Colour" settings in WSJT-X
 - Those with the Best S/N ratio
 - DX stations only
 - DX stations preferred
 - Those at the greatest distance
 - Wanted stations are preferred (according to wanted lists of the Helper)
 - Wanted stations only (according to wanted lists of the Helper)
 - Stations, countries and continents can be blocked using "exclude lists"
- Band-Hopping (automatic band switching with time control)
 - Daily schedules, each with to time periods
 - Band, FT-mode, and filter strategy can be specified.
 - Convenient input of all parameters
- Easy installation of the Helper
- Opens the QRZ.com page of the QSO-partner when a QSO starts
- Band and FT-Mode are controlled by the Helper
- Protection against malicious attacks or malfunctioning of stations



Main Window of the FT8-Helper

The Main Window - human interface:

Own call / grid:

Your own call sign and QRA locator, must be entered into the configuration using the “Edit config” dialog.

Edit config:

Opens the configuration dialog of the Helper. The configuration data is stored in the “config.txt” file located in the installation folder. See more details below in the text.

About:

Displays the current version number and copyright information (Freeware)

Auto Frq:

If “Auto-Frq” is selected in “CQ” or “MIX”-mode, the transit frequency is automatic set to a free frequency between 500 and 2800Hz. The transmit frequencies of the stations received during the last four consecutive cycles are aggregated and the largest gap between these stations is selected as transmit frequency.

Important notes:

- If “Auto Freq” is to be used, the entire “Wide-Graph” window must be visible on the screen and its horizontal size must not be altered. The frequency range between 500 -2800Hz must be visible. Please refer to the required settings below in the chapter “Waterfall Diagram Settings”.
- WSJT-X 3.0.1 itself offers various filtering functions, as well as the option to display only stations calling CQ or transmitting 73. If “Auto-Freq” is to be used, all filters must be disabled!

Skip Tx-1:

In "S/P" mode, the Helper calls stations using a signal report instead of a QRA locator. This allows the QSO to be somewhat shortened. However this is not recommended on the 6m band, as the QRA locator is of great importance on this band.

Keep Even/Odd:

If this function selected, the Helper calls CQ only during the selected even or odd period. If neither option is selected, the CQ is transmitted at the next available period. This setting is inactive in S/P-mode. This function is most commonly used in 6m DX operations.

Free channels / Use:

This field displays the number of free channels and the transmission frequency selected by the "Auto Freq" function of the Helper. If "Auto-Freq" is not active, or if no free frequency could be found, the fields will display "—" and "-----"

Decode Delay (Lag time):

In most cases WSJT-X cannot decode all messages within 15/7,5 sec. This value indicates how long WSJT-X requires for decoding the messages during the subsequent cycle. The decoding depth in WSJT-X should be configured such that this value remains below 500ms. If it exceeds 1000 msec, there is a risk that transmitted messages may not be decoded by the receiving station.

Mode Sel:

"Mode-Sel" is used to select the desired mode of operation: S/P CQ, XCHG, or MIX.

When starting S/P mode, the Helper disables "Hold Tx Freq" in WSJT-X, conversely, in CQ and MIX modes, "Hold Tx Freq" is enabled. Although not recommended, it can be changed after startup.

CQ-Mode:

In CQ mode, WSJT-X continuously transmits CQ calls. QSOs are conducted automatically. The number of unanswered CQ calls can be configured in the Helper's configuration menu. Once the maximum number of CQ calls has been reached, the Helper enters "Sleep" mode for a programmable number of cycles.

S/P-Mode:

In "Search and Pounce" mode, the Helper selects stations calling CQ. Which stations are called first, depends on the configured "Auto-Strategy" (see below). When the function 73=CQ is enabled, stations sending "73" or "RR73" will also be called, provided that the Helper has previously received "CQ" from that station in the background.

XCHG Mode:

In XCHG-Mode, the system automatically switches between "CQ" and "S/P" modes. It always starts in S/P mode and switches to CQ mode after a programmable number of unsuccessful (broken) QSOs.

The maximum number of QSO attempts can be configured in the Helper's "Edit-Config" dialog:

- "Max. broken QSOs in CQ-cycle" for CQ operation,
- "Max. broken QSOs in S/P cycle" for S/P operation.

The "Progress" field displays both the current number and the maximum number of attempts.

MIX- Mode:

In MIX mode, the Helper calls CQ and waits for responses. If no one responds, the Helper calls stations that are calling CQ, in accordance with the configured Auto Strategy. Upon completion of the QSO, it resumes calling CQ. If, once again, no one responds, it searches for a CQ-calling station again.

START and STOP:

Starts and stops operation in the pre-selected mode (S/P CQ, etc.). Pressing the "STOP" button once initiates the shutdown sequence, but waits the current QSO is over. The field shows "Wait Down" in yellow. Pressing the "STOP" button a second time terminates all activities immediately. The field below displays the current operational status.

Important Note: After startup, the Helper disables "Auto Seq" in WSJT-X. The Helper uses its own auto-sequencing procedure. Auto-Seq must not be manually re-enabled during operation.

QSO With:

This field displays the call sign of the current QSO. The color corresponds to the priority color shown in the "Band activity" window. Logged stations are displayed in green, unlogged stations in red. Incoming calls are orange.

Progress:

This field displays the progress in XCHG mode and during "band hopping". :

73=CQ:

If this function is selected in S/P mode, not only stations calling CQ, but also stations sending RR73 or 73, will be called, provided that the Helper has previously received "CQ" from that station in the background.

If you set "Hold-Tx-Freq" manually in S/P mode and you are on a free frequency, the Helper calls stations even if no CQ was previously received.

Repeat RR73:

If this function is selected, the Helper repeats the RR73 message if no concluding 73 is received from the other station.

QRZ.com:

Opens the QRZ page for the station with which the QSO was initiated. This uses the browser currently set as the default in Windows.

Auto-Seq. Strategies:

The selected Strategy can only be changed if the running operation is stopped.

- **Best-S/N:**
The CQ-calling station with the highest priority is called. The lowest priority is "New Call in Band/Mode," and the highest priority is "New Continent." Only the colours selected in the "Colours" dialog in WSJT-X are taken into account. If multiple stations with the same priority are received simultaneously, the station with the best signal-to-noise ratio is called.
- **DX-only:**
Only DX stations are called. Stations are considered DX stations if the distance calculated from the QRA locator is greater than the minimum DX distance specified in the "Edit Config" dialog.
- **Prefer-DX:**
If no DX stations are being received, other stations are called using the "Best S/N" method.
- **Most-distance:**
Stations located farthest away are called first. The distance is calculated by the QRA locator of the station.
- **Prefer-wanted:**
In the "Edit Config" dialog, you can specify a list of preferred (Wanted) stations. You can specify call signs, prefixes, DXCCs, QRA locators, and continents. Stations that match the "Wanted" lists are called first. (See also the "Edit Config" dialog below.)

- **Only-wanted:**

Only stations that match the “Wanted” entries are called. Stations with a priority of “New-DXCC on Band” or higher are always considered “Wanted,” provided that the corresponding checkbox has been selected in the “Colours” dialog in WSJTX.

- **DX-Chasing:**

DX chasing was developed for “hunting” rare stations. The desired station should be entered in the “Prompt Calls” field or into the “Wanted Calls” field in the Helper configuration. The Helper will call the station as long it can be received.

Important note: DX-Chasing should always be operated with “Hold Tx Freq” enabled. It is also recommended to set the TX frequency manually and search for a new frequency from time to time.

BAND:

Starting with version 3 of WSJT-X, a new option for band switching has been implemented. To use this feature, check the “Band Buttons” box in the “View” menu of WSJT-X. The Helper now controls the bands via these buttons. Upon startup, the Helper checks whether the “Band Buttons” are enabled.

FT-Mode:

The selection of FT modes has also been updated in WSJT-X Version 3. The Helper switches between FT-4 and FT-8 using these new buttons.

Prompt Calls:

Enter the desired stations here, separated by spaces. This list is not saved when you exit the Helper. Call signs can be entered during without stopping the operation, although this is somewhat difficult especially on the FT-4 because the helper keeps taking the focus away. Worked stations are automatically removed from the list.

The „Edit Config“ Dialog

- My-Call ; your station`s call-sign
- My-Grid ; the four character QRA locator (grid) of your own station
- MY-Area ; Your own country and continent, also TEST SOTA or POTA are accepted.
Directed CQ calls only considered, if the direction is in this list. No entry need for DX.
- Macro Delay ; Some computers require a value between 10 and 50 here
- Min. DX distance ; The desired minimum DX distance; Typically 2000 Km
- Km/Mile ; The DX distance in kilometres or miles
- Max. audio freq. ; Stations calling above the specified frequency will be ignored.
- Min. S/N level ; Stations with an S/N value below this value are ignored
- Max. CALL repeats ; Maximum number of call repeats when calling stations
- Max. CQ/Sleep ; Maximum number of CQs without answer before entering the “sleep” period.
- Max. repeats in QSO ; Maximum retries for report exchange (QSO not yet logged)
- Max. repeats at QSO-end ; Maximum retries of RR73 and 73 at QSO-end (QSO already logged)
- Max. broken QSOs in CQ ; Maximum number of broken QSOs in CQ-Mode (for XCHG and Band Hopping)
- Max. broken QSOs in S/P ; Maximum number of broken QSOs in S/P-Mode (for XCHG and Band Hopping)
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- Oldest accepted call ; Stations that call us during a QSO are stored and recalled at the end of the current QSO, but only if the call is no older than the value specified here. Typical values are 1–2 minutes..
- Logging level ; Log is written into the file "\\log\\log-<YYMMDD.txt" in the installation directory;
 - Logging level: NONE ; No logging file written
 - Logging Level: LOG ; Only the content of the log-window of the FT8-Helper is logged
 - Logging level: MSG ; Additionally, all received and transmitted messages are logged
 - Logging level: EXPERT ; Additionally, information for software debugging are logged
- Excluded DXCCs ; List of DXCC countries and continents you **don't want** to work. Valid DXCC prefixes as specified in WSJTX and following continents are accepted: *EU, *AF, *AS, *NA, *SA, and *OC. Placing a "#" before the DXCC prefix will prevent stations from that country from being actively called, but calls from that country will still be accepted.
- Excluded Calls ; List of stations and prefixes you **don't want** to work. A "JA1" it applies only to call signs whose first 3 characters are "JA1". Invalid countries, such as "D0," can only be entered here.
- Wanted-Grids ; List of QRA locators, separated by spaces, where you would like to work. Two or four characters are accepted.
- Wanted DXCCs ; List of DXCC countries and continents you would like to work. The DXCC prefixes and continents can be entered in a similar way to the "Excluded DXCCs" field.
- Wanted Calls ; List of call signs or prefixes you want to work. Similar to "Excluded Calls," you can enter call signs and prefixes. Another option is to enter special stations that all share the same suffix, such as POTA, SOTA, or VOTA". For example *POTA would call all stations whose call sign end with "POTA".

Band-Hopping:

"Band Hopping" allows for the automatic switching of the operating frequency, FT mode, and QSO strategy according to a programmable schedule.

The sequence program can be entered line by line. Select the desired band, time, FT mode, and strategy, then press the "Add Line" button. The line is added to the program list. Time intervals within a single line must not overlap. If only one time interval is required, the second field must be left blank. Times from 0000 to 2400 are accepted.

When "Band Hopping" is started, a window displaying the sequence program opens. Bands that can be activated at the time of startup are marked with a "+" at the beginning of the line. After pressing "OK," the program marked with "->" begins.

Important Notes:

- The times are local computer times, not UTC!!
- The lines can be edited, copied, or deleted later.
- Unneeded lines can be excluded by placing a "#" character at the beginning of the line.

QSOs:

- The first number shows the number of QSOs logged during the current CQ or S/P session.
- The second number shows the total number of QSOs logged since the helper was started.

Broken (QSOs):

The number of QSOs started but not completed since the Helper was launched

Rep Diff:

The average difference between transmitted and received S/N reports. A positive value indicates that transmit power could be reduced.

The Require Settings

Settings in Windows 10/11:

- Use the maximum resolution of your screen. The minimum resolution is 1360 x 768 pixels. If the vertical resolution is less than 1050 lines, the helper will start in “low-resolution” mode. In this mode, WSJT-X starts up slightly smaller, and WSJT-X overlaps the Wide-Graph window. The size of WSJT-X must not be changed, however it can be moved.
- Set the windows system display scaling to 100%
- Use only the default WIN-11 window manager.
- Some remote programs may cause unexpected impact on the functionality

Required settings in WSJTX Main Window

- As a general rule, you should not use any new WSJT-X features that you do not need.
- Do not activate “Use Dark Style” in the View menu..
- The settings in the “Decode” menu should be adjusted according to your PC's performance. The “Decode” phase should be completed no later than half a second after the next cycle starts. Monitor the “Decode Delay” display in the FT8 Helper.
- “CQ/73” must not be selected and all filters in WSJTX must be disabled if “Auto-Frq” is to be used.
- The “Menu” checkbox must be activated. The Helper will alert you if the setting hasn't been configured.
- In the “DX field”, tab “1” must be selected so that the TX messages TX-1 through TX-6 are accessible.
- “Tx even/1st” and “Hold Tx Freq” are controlled by the helper.
- “Enable TX”, “Halt-TX”, and “Log QSO” are controlled by the helper.
- “Auto Seq” is disabled at start-up and must not be enabled manually.
- The sizes of the WSJT-X and Wide-Graph set by the Helper at run-up must not be changed.
- The windows may be moved, but must always remain entirely on the desktop.
- The vertical split is set automatically on run-of the Helper and must not be changed. However, it should be roughly centered before starting the Helper.
- The default setting for “Font” is: MS Shell Dlg 2, Font style: Standard, Size: 8,
- The default setting for “Decoded Text Font”: Courier or Courier New, Font style: Standard, Size: 10,
- The “Band Buttons” implemented for WSJT-X Version 3 must be enabled. The Helper checks whether the buttons are enabled before starting and alert you if the Band-Buttons are not available.

Settings in WSJT-X "General":

The screenshot shows the 'Settings' dialog box for WSJT-X, with the 'General' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar are tabs for 'General', 'Radio', 'Audio', 'Tx Macros', 'Reporting', 'Frequencies', 'Colors', 'Advanced', 'Alerts', and 'Filters'. The 'General' tab contains three main sections: 'Station Details', 'Display', and 'Behavior'. The 'Station Details' section has fields for 'My Call' (DG5LP), 'My Grid' (JN48f), an 'AutoGrid' checkbox, an 'IARU Region' dropdown (All), and a 'Message generation for type 2 compound callsign holders' dropdown (Full call in Tx3). The 'Display' section has checkboxes for 'Start new period decodes at top', 'Blank line between decoding periods', 'Show blank line with time stamp', 'Display distance in miles', 'Highlight DX Call in message', 'Tx messages to Rx frequency window', 'Highlight DX Grid in message', 'Show DXCC, grid, and worked-before status', 'Show principal prefix instead of country name', and 'Show DXCC for all messages'. There are also buttons for 'Font...' and 'Decoded Text Font...'. The 'Behavior' section has checkboxes for 'Monitor off at startup', 'Enable VHF and submode features', 'MSK144/Q65: Tx until 73 is received', 'Monitor returns to last used frequency', 'Allow Tx frequency changes while transmitting', 'Auto-open/close Astronomical data window', 'Double-click on call sets Tx enable', 'Decode after EME delay', 'kHz entry without k (restart required)', 'Disable Tx after sending 73', 'Single decode', 'Turn progress bar red while transmitting', 'Calling CQ forces Call 1st', 'CW ID after 73', 'ID Interval' (0), 'Alternate F1-F6 bindings', 'Tune watchdog' (90 seconds), and 'Tx watchdog' (6 minutes). The 'Additional features' section has checkboxes for 'Map Grid locator to U.S. State', 'Erase decodes after band change', 'Alternate Erase button behavior', 'Show U.S. States in messages with grid', 'Set Rx frequency to Tx frequency after QSO', 'Enable Wait features', 'Clear DX Call after QSO', 'Clear DX Grid after QSO', 'Less button coloring (restart required)', 'Show distance in messages with grid', 'Show azimuth in messages with grid', and 'Align DXCC/distance/azimuth' (2, 0). At the bottom right are 'OK' and 'Cancel' buttons.

Settings

General Radio Audio Tx Macros Reporting Frequencies Colors Advanced Alerts Filters

Station Details

My Call: DG5LP My Grid: JN48f ☐ AutoGrid IARU Region: All

Message generation for type 2 compound callsign holders: Full call in Tx3

Display

☐ Start new period decodes at top ☒ Blank line between decoding periods ☒ Show blank line with time stamp

☐ Display distance in miles ☒ Highlight DX Call in message Font...

☒ Tx messages to Rx frequency window ☐ Highlight DX Grid in message Decoded Text Font...

☒ Show DXCC, grid, and worked-before status ☒ Show principal prefix instead of country name ☐ Show DXCC for all messages

Behavior

☐ Monitor off at startup ☐ Enable VHF and submode features ☐ MSK144/Q65: Tx until 73 is received

☐ Monitor returns to last used frequency ☐ Allow Tx frequency changes while transmitting ☐ Auto-open/close Astronomical data window

☐ Double-click on call sets Tx enable ☐ Decode after EME delay ☐ kHz entry without k (restart required)

☒ Disable Tx after sending 73 ☐ Single decode ☒ Turn progress bar red while transmitting

☐ Calling CQ forces Call 1st ☐ CW ID after 73 ID Interval: 0

☐ Alternate F1-F6 bindings ☒ Tune watchdog 90 seconds Tx watchdog: 6 minutes

Additional features

☐ Map Grid locator to U.S. State ☐ Erase decodes after band change ☒ Alternate Erase button behavior

☐ Show U.S. States in messages with grid ☐ Set Rx frequency to Tx frequency after QSO ☒ Enable Wait features

☐ Clear DX Call after QSO ☐ Clear DX Grid after QSO ☐ Less button coloring (restart required)

☐ Show distance in messages with grid ☐ Show azimuth in messages with grid ☒ Align DXCC/distance/azimuth 2 0

OK Cancel

The Helper was tested with the settings about, however not using this setting may have impact on the correct function. Of course, you can change country-specific settings, such as:

- Show country name instead of DXCC prefix
- Show US states
- Use miles

Settings in WSJT-X "Reporting":

All logging checkboxes except "Log automatically" must remain unchecked. "Prompt me to log QSO" **must not** be selected. These settings are very important!

Settings in WSJT-X "Frequencies":

Be sure that for both FT4 and FT8 the correct frequency offsets are programmed.

Settings in WSJT-X "Colours":

- The "Highlighting" colour "CQ in message" (green) must be selected.
- "Transmitted message" (yellow) should also be selected.
- At least one additional colour marker must be selected, typically "New Call on Band."
- Other colour markers can be selected as needed.
- In the lower section, "Highlight messages with 73 or RR7" should be enabled.
- The colour scheme **must not** be changed. The helper only recognizes the original colours.

Correct settings of the Wide-Graph window:

- Bins/Pixel = 3
- Start 500 Hz
- All frequencies from 500 Hz to 2850 Hz must be visible.
- This setting is mandatory only if the “Auto-Frq” feature should be used.

Installation and Preparation before first use

Download the program (see link at www.github.com/dg5lp and copy it to any directory. Launch WSJT-X, then the FT8 Helper. Upon first launch, call sign and QRA locator fields will be empty. Open the “Edit Config” dialog and enter your personal data; “My-Call,” “My Grid,” and “My Locations”. Please don’t forget to click “OK” to save the configuration. The FT8 Helper is now ready for use. Start “CQ” or “S/P” operation.

Bug reports and observations to:

dg5lp@darc.de or via github.